Retrieval-Conditional Parsing Score (RCPS): Choosing Document Parsers by Retrieval, Not by Appearance

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Abstract

Document RAG retrieves from parser outputs, yet parsers are commonly selected by intrinsic measures of transcription fidelity and boundary quality rather than by retrieval performance. We introduce the Retrieval-Conditional Parsing Score (RCPS), a training-free protocol that ranks parser–chunker combinations on a fixed held-out Q–A retrieval probe. RCPS averages mean reciprocal rank (MRR) across specified retrievers and retrieval depths, using the same format-normalised exact-span relevance rule for every candidate. On KoGovDoc-RAG, table-enabled MinerU-on has a higher Boundary Clarity score than our production parser, Prod (0.713 versus 0.610). Yet its Hit@1 is 42.6 percentage points lower (0.123 versus 0.549). Across five parsers evaluated on all 294 pages, RCPS ranges from 0.137 to 0.584. With Prod fixed, the four chunkers score between 0.535 and 0.593. A retriever-free coverage diagnostic finds no normalised exact match for 20.2% of reference spans in Prod output. No tested chunker splits more than 2.3%. On source-aligned OHR-Bench data, semantic noise lowers retrieval while Boundary Clarity remains stable or changes non-monotonically. In a secondary parser-training study, two direct preference optimization checkpoints improve Hit@5 by only 0.95 and 1.15 percentage points over Prod on an audited compatibility subset; a matched edit-distance control performs similarly. The workflow therefore selects with RCPS, diagnoses with coverage, and trains only if needed. We release the 663-Q–A KoGovDoc-RAG probe and an RCPS reference implementation.

1 Introduction

A document RAG pipeline uses a parser to convert pages into text, a chunker to divide that text into retrievable units, and a retriever to rank those units for a query. Document-parsing benchmarks commonly rank systems using intrinsic output-fidelity metrics (Ouyang et al., 2025). We additionally consider Boundary Clarity (BC), an intrinsic chunk-boundary metric (Zhao et al., 2025a). These measures evaluate parser output in isolation. They do not show whether the parser preserves evidence in a form that the downstream retriever can find. If the parser omits or corrupts answer-bearing evidence, downstream chunking and retrieval cannot restore the exact span.

The mismatch is large in our deployment pool. Table-enabled MinerU-on (Niu et al., 2026) has higher BC than our production parser, Prod (0.713 versus 0.610), but much lower Hit@1 (0.123 versus 0.549). Prod's Hit@1 advantage is 42.6 percentage points. A source-aligned OHR-Bench perturbation study (Zhang et al., 2025a) provides a controlled check of the same mismatch. Semantic corruption lowers retrieval, while BC remains stable or changes non-monotonically. A well-structured parse can therefore still be poor input to a retriever.

We address the deployment decision before attempting parser training. The Retrieval-Conditional Parsing Score (RCPS) evaluates every candidate parser–chunker combination on the same held-out Q–A probe. It averages mean reciprocal rank (MRR) over specified retrievers and retrieval depths under one relevance rule. RCPS is deliberately a simple protocol, not a new similarity function. Across five complete parser outputs, RCPS ranges from 0.137 to 0.584. With Prod fixed as the parser, scores for four chunkers range from 0.535 to 0.593.

A retrieval score identifies the better pipeline but not the faulty layer. We therefore

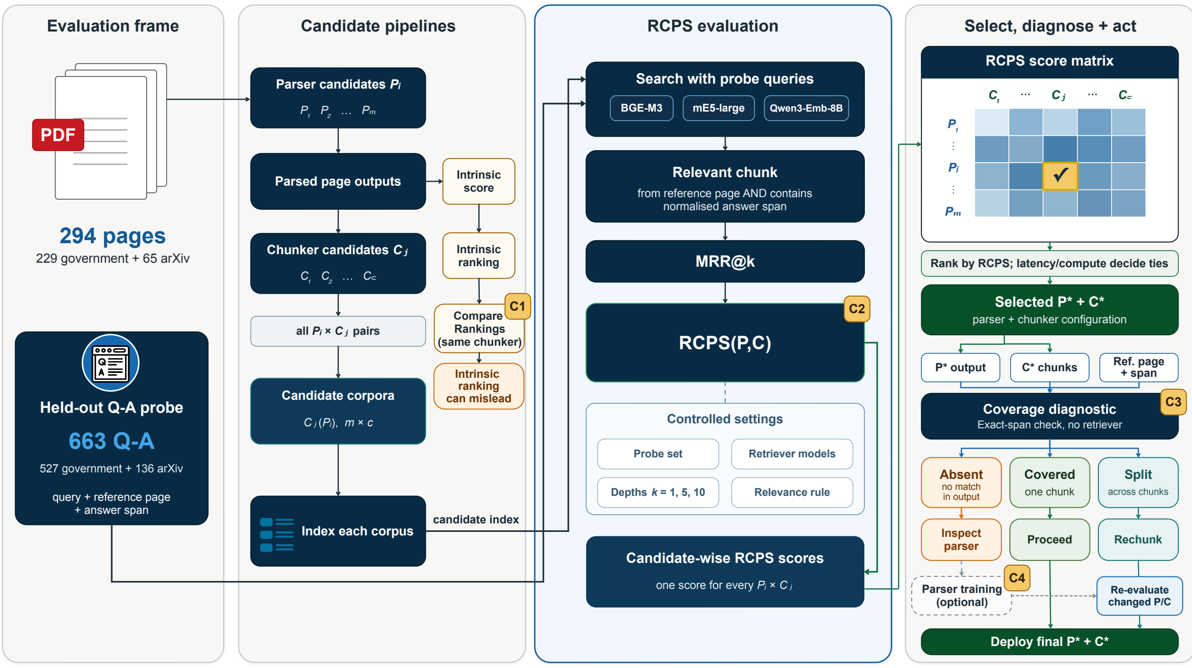

Figure 1: **RCPS workflow.** A held-out Q–A probe evaluates parser–chunker candidates. Intrinsic and RCPS parser rankings can disagree (C1). RCPS ranks candidates across fixed retrievers and retrieval depths (C2). Coverage separates spans absent from parser output from those split across chunk boundaries (C3). Parser training is reserved for unresolved parser-side failures before changed configurations are re-evaluated (C4).

add a retriever-free coverage diagnostic. It distinguishes reference spans that are absent from the normalised parser output from spans that exist in the output but cross chunk boundaries. For Prod, 20.2% of reference spans have no normalised exact match before chunking. No tested chunker splits more than 2.3%. This points first to parser-output inspection rather than further chunker tuning.

We run two secondary experiments after selection and diagnosis. In a three-parser end-to-end check, the RCPS top choice, Prod, also has the highest answer accuracy (72.5% versus 23.8% and 20.5%). The lower pair reverses, so the experiment supports only the top choice. We also train the parser with retrieval-oriented feedback. Two direct preference-optimization checkpoints have Hit@5 estimates only about one percentage point above Prod on a separate, audited OHR-Bench compatibility subset. A matched edit-distance control performs similarly, so the comparison does not isolate a benefit from retrieval reward.

Figure 1 summarises the resulting workflow: select with RCPS, diagnose parser- versus chunker-side loss with coverage, and consider parser training only afterward.

Contributions. We first show that intrinsic metrics can misrank parsers (C1). Our primary contribution is RCPS, a training-free protocol for selecting parsers and chunkers by retrieval (C2). We then introduce coverage as a diagnostic for separating parser-side absence from chunker-side splitting (C3). Finally, we report the small and uncertain retrieval differences associated with parser training (C4). The main scope is answer-span retrievability. Appendix B provides the three-parser end-to-end check.

We release the frozen 663-Q–A KoGovDoc-RAG probe and a reference implementation of RCPS.[1] The selection experiment indexes 294 pages. Appendix H records the remaining release and clean-rerun gaps.

2 Related Work

Parser and document-retrieval evaluation. Parsing benchmarks such as OmniDocBench rank systems by intrinsic fidelity (Ouyang et al., 2025). These scores measure

[1] https://github.com/wigtn/WigtnOCR-RADP

transcription quality but do not directly evaluate the corpus seen by a retriever. OHR-Bench (Zhang et al., 2025a) traces OCR errors through RAG, while EnterpriseDocBench (Singh and Raj, 2026) and concurrent studies (Sun et al., 2026; Song, 2026) report related gaps between parsing quality and retrieval. Vision-based retrieval (Faysse et al., 2025) avoids text parsing altogether. We instead keep the text-RAG pipeline and use retrieval to select its parser and chunker. Our coverage diagnostic then separates reference spans absent from parser output from spans split across chunk boundaries.

Retrieval-oriented training. Prior work improves chunking or chunk representations (Günther et al., 2024; Duarte et al., 2024; Zhao et al., 2024, 2025a; Singh et al., 2024), document embeddings (Conti et al., 2025; Zhao et al., 2025b), retrievers, or readers (Nguyen et al., 2024; Chia et al., 2025; Yan et al., 2025). Parser training has instead used edit-distance and layout rewards (Wang et al., 2026a) or query guidance (Wang et al., 2026b). Our secondary study applies retrieval feedback to the parser and compares it with a matched edit-distance fidelity control. The selection and coverage contributions require no training.

3 Two Tools: Select and Diagnose

In our pipeline, a parser converts each page image into Markdown. A chunker divides that transcription into retrievable units, and a retriever ranks those units for a query. Appendix C specifies the parser input and output. The two tools below answer different questions. RCPS selects the parser–chunker combination. Coverage diagnoses whether an exact span is absent before chunking or split across chunk boundaries.

3.1 RCPS **Selects by Retrieval**

RCPS evaluates every candidate on the same held-out Q–A probe. For each query, MRR@k is the reciprocal rank of the first relevant chunk if that chunk appears in the top k, and zero otherwise. It then averages over queries. A chunk is relevant only when it comes from the reference page and contains the reference answer after shared whitespace and Markdown normalisation. Every candidate uses this same

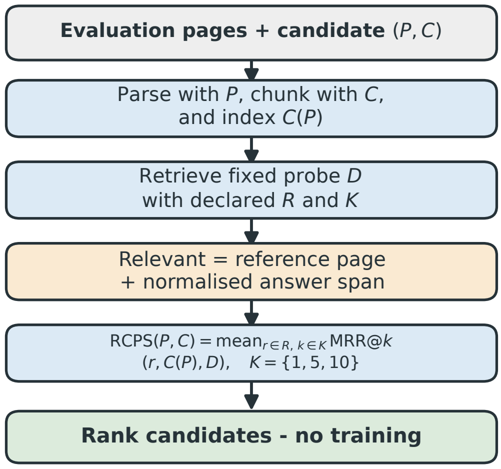

Figure 2: RCPS compares candidates on the same held-out Q–A probe. It averages standard MRR over specified retrievers and retrieval depths. A hit must come from the reference page and contain the normalised reference answer. Use a singleton retriever set when deployment is fixed, or a specified set when it is undecided. Evaluation itself requires no training.

rule on its own parsed-and-chunked corpus. Equation 1 defines the RCPS *score*. The shared evaluation procedure is the RCPS *protocol*.

Let P denote a parser, C a chunker, and $D = \{(q_i, a_i, \text{page}_i)\}$ the held-out Q–A probe. Let R be the retriever set and K the evaluated retrieval depths. RCPS averages MRR over all retriever–depth pairs:

$$\text{RCPS}(P, C) = \frac{1}{|R|\,|K|} \sum_{r \in R} \sum_{k \in K} \text{MRR@}k(r, C(P), D). \tag{1}$$

Here, $C(P)$ is the corpus produced by parser P and chunker C. We use BGE-M3 (Chen et al., 2024), multilingual-e5-large (Wang et al., 2024), and Qwen3-Embedding-8B (Zhang et al., 2025b) for R, with $K = \{1, 5, 10\}$. Averaging across these retrievers and depths reduces dependence on any one setting. Figure 2 shows the workflow.

Execution. Build a representative held-out probe whose answers occur verbatim on their reference pages. Our probe is model-generated. An LLM-assisted quality check accepted 94/100 sampled pairs; Section 4.1 reports its criteria. Parse each page once per parser, then chunk, index, and search every candidate corpus. With m parsers, c chunkers, and $|R|$ retrievers, this requires m parsing runs and $mc|R|$ retrieval evaluations.

Interpretation. The protocol needs neither training nor manual chunk labels because relevance follows from the answer span and source page. Fixing C ranks parsers. Fixing P ranks chunkers. The ranking is relative to the candidate pool and probe.

3.2 Coverage Separates Absence from Splitting

Coverage does not score a retriever. With parser output fixed, a reference span is *covered* when it appears within one chunk. It is *split* when it exists in the page output but crosses chunk boundaries. It is *absent* when the normalised page output contains no exact match. Overlap may repair a split. Rechunking alone cannot restore an absent exact span. These are operational labels: *absent* does not prove that the meaning is missing.

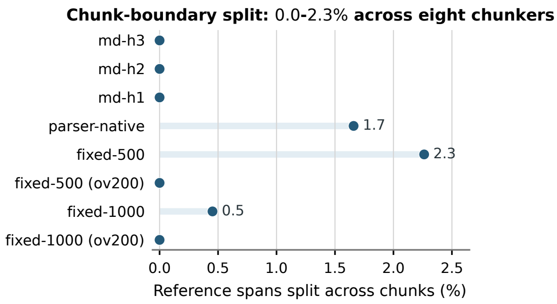

Figure 3: Coverage with Prod fixed (294 pages, 663 Q–A; no retriever). Absence is measured before chunking and stays at 20.2%. Changing the chunker affects only splitting, which reaches at most 2.3%.

4 Experiments

4.1 Evaluation Frames and Setup

Evaluation frames. We keep five evaluation frames separate. The parser- and chunker-selection experiments evaluate all 663 KoGovDoc-RAG Q–A against the same 294-page corpus: 229 government pages and 65 arXiv pages. The answers occur on 242 of those pages; the other 52 serve as Q–A-free distractors. Training and mechanism analyses index only those 242 evidence-bearing pages. The pre-specified pilot uses 73 pages and 202 Q–A. For perturbation analysis, OHR-Bench (Zhang et al., 2025a) provides 1,043 aligned Law–Manual Q–A. Training evaluation uses an aligned OHR-Bench compatibility subset (2,036 Q–A, six domains; Appendix A). Results from different frames are not directly comparable.

Data and leakage control. Qwen3-VL-30B produced the pseudo-reference Markdown, which we manually de-noised. `gpt-5.4-2026-03-05` generated the Q–A. A separate LLM-assisted check assessed question clarity, answer correctness, and whether the answer span appears in the reference context. It accepted 94/100 sampled pairs. **Prod** is our Qwen3-VL-2B parser (Bai et al., 2025) fine-tuned for Korean documents. It serves as both a deployment candidate and the training baseline. Training pages are disjoint from every evaluation frame.

For MinerU, *MinerU-off* is the originally submitted output, generated with table recognition disabled. *MinerU-on* is the later audited, table-enabled output used in the deployment comparison and four-parser BC correlation. We retain MinerU-off only as a separately labelled diagnostic. The two runs also differ in software and retrieval environments. Their difference is therefore not a controlled table-recognition ablation.

4.2 Boundary Clarity Can Misrank Parsers in This Pool (C1)

RCPS and Boundary Clarity rank parsers differently in this candidate pool (Figure 4; Table 1). MinerU-on has the highest measured BC among complete outputs (0.713) but the lowest RCPS (0.137). The 30B teacher and Prod have lower BC yet the highest RCPS values, 0.584 and 0.583. Across all five complete 294-page outputs, RCPS spans 0.137–0.584. PaddleOCR's parser-native output has no adjacent chunk boundaries, so BC is undefined for that configuration.

Among the four complete outputs with defined BC, the Pearson correlation with RCPS is $r = -0.74$. Adding Marker's partial 38-page output gives $r = -0.83$ $(n = 5)$. Marker itself scores 0.073 RCPS and is excluded from complete-output comparisons. These small-sample correlations describe this pool only.

Law–Manual perturbations (1,043 Q–A). On the aligned OHR-Bench subset, se-

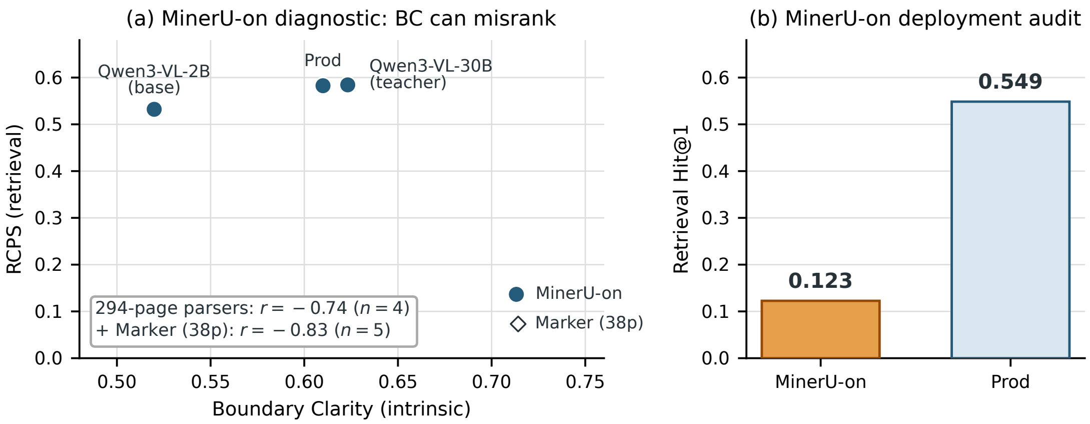

Figure 4: In this candidate pool, higher Boundary Clarity (BC) does not imply better retrieval. (a) Among four 294-page parsers with defined BC, the correlation with RCPS is $r = -0.74$. Adding Marker's partial 38-page output gives $r = -0.83$ ($n = 5$). MinerU-on is used in this diagnostic. (b) Prod exceeds MinerU-on Hit@1 by 42.6 points (0.549 vs. 0.123; 4.47×).

mantic corruption lowers retrieval without a consistent BC response. From mild to severe noise, RCPS falls 0.476 → 0.265 for MinerU, 0.534 → 0.497 for Qwen2.5-VL, and 0.461 → 0.298 for GOT. The corresponding BC values behave differently: MinerU stays within 0.628–0.651, Qwen2.5-VL remains near 0.563, and GOT rises 0.586 → 0.624. These twelve variants are dependent perturbations of three outputs, not independent parsers (Appendix A), and do not establish broad domain generality.

4.3 RCPS **Ranks Parsers and Chunkers (C2)**

Table 1 applies the same probe, retrievers, depths, and relevance rule to every parser. The companion chunker comparison fixes Prod and changes only the chunker.

Selecting parsers. The 30B teacher ranks first at 0.584 RCPS, followed by Prod at 0.583. Prod nevertheless has slightly higher Hit@1 (0.549 versus 0.545). The point estimates are near-tied; in the probe bootstrap below, the teacher remains above Prod in only 62.5% of draws. Latency and compute may therefore decide between them. MinerU-on reaches only 0.137 RCPS and 0.123 Hit@1. Prod's Hit@1 is 4.47× as high.

Selecting chunkers. With Prod fixed, the point-estimate order is md-h3 (0.593) > parser-native (0.583) > LumberChunker (Duarte et al., 2024) (0.557) > fixed-500 (0.535). The chunker range is 0.058. This is much smaller than the 0.447 range across the heterogeneous parser pool and close to the 0.052 range among its three vision–language parsers.

Retriever, depth, and probe robustness. When deployment fixes a retriever, evaluate with that retriever. Otherwise, averaging reduces dependence on one model. In the stored MinerU-off aggregate grid, averaging three retrievers at MRR@10 alone preserves both the five-parser and four-chunker orders ($\tau = 1.0$). We also draw 1,000 fixed-seed subsets of 500/663 Q–A without replacement

Parser	BC	CS	RCPS	Hit@1
Deployment comparison (294 pages)				
Qwen3-VL-30B (teacher)	0.623	3.38	**0.584**	0.545
Prod (ours, 2B)	0.610	3.07	0.583	0.549
Qwen3-VL-2B (base)	0.520	3.74	0.532	0.500
PaddleOCR	—	3.46	0.140	0.125
MinerU-on	**0.713**	—	0.137	0.123
Partial output				
Marker (38p)	0.717	3.41	0.073	0.068

Table 1: RCPS ranks the 30B teacher and Prod at the top of the complete-output parser pool. Scores average three retrievers and $k \in \{1, 5, 10\}$. BC is Boundary Clarity, for which higher is better. CS is Chunk Stickiness, for which lower is better. PaddleOCR has no adjacent parser-native boundaries from which to compute BC. CS was not recomputed for MinerU-on after the audited table-enabled rerun. The four-parser deployment correlation uses MinerU-on. Marker is excluded from complete-output comparisons.

from the aligned 294-page full grid. The six-configuration parser order has mean Kendall $\tau_a = 0.902$; its exact order is unchanged in 39.4% of draws because the teacher–Prod and PaddleOCR–MinerU-on pairs are near-tied. Prod remains above Base, MinerU-off, PaddleOCR, and MinerU-on in every draw. The four-chunker order is unchanged in 96.1% of draws, and md-h3 remains above parser-native in 96.5%. Replacing format-normalised relevance with raw substring matching lowers RCPS by 0.024–0.041 without changing either full order. A single-retriever ablation and its limits appear in Appendix H.

4.4 Coverage Finds Pre-Chunking Absence (C3)

The retriever-free diagnostic uses all 294 pages. For Prod, 134/663 reference spans (20.2%) have no normalised exact match before chunking. No tested chunker splits more than 15/663 spans (2.3%) (Figure 3). Thus, rechunking alone cannot restore exact-span retrievability for roughly one in five reference answers.

Under the same matcher, MinerU-on's absent rate is 66.1%, 45.9 points above Prod. Because absence is matcher-defined, Appendix C checks the label with tolerant matchers, a cross-family judge, and human verification.

4.5 Parser Training Is a Secondary, Bounded Lever (C4)

After selection and diagnosis, we refer to retrieval-oriented parser training as Retrieval-Aware Document Parsing (RADP). RADP-aux adds a contrastive hidden-state objective, while RADP-DPO turns page-local retrieval scores into preference pairs (Rafailov et al., 2023). RADP-Distill replaces the retrieval reward with edit distance as a fidelity control. These methods change the parser. RCPS itself does not. Appendix E gives the objectives and training setup.

Neither RADP-aux nor RADP-DPO meets the pre-specified pilot target: a 5-percentage-point RCPS gain with a 95% CI lower bound above zero. On the separate 2,036-Q–A OHR-Bench compatibility subset, R2, R3, and RADP-Distill exceed Prod Hit@5 by only 0.95, 1.15, and 1.36 points. The paired Distill-minus-DPO intervals include zero for both R2 and R3, so these data do not identify a benefit from the retrieval reward over the edit-distance control. These post-audit estimates are neither a full v2 rerun nor a pre-specified confirmation. Appendix F reports the full results and the SimPO control (Meng et al., 2024).

5 Discussion and Conclusion

The main deployment lesson is to select before optimising. In our candidate pool, choosing Prod rather than MinerU-on raises Hit@1 by 42.6 percentage points. On a different OHR-Bench frame, the audited parser-training checkpoints exceed Prod Hit@5 by about one percentage point. The magnitudes are not directly comparable, but they place candidate selection before further training.

The diagnostics explain how to act on that selection. High BC does not guarantee retrievable content: MinerU-on combines BC 0.713 with 66.1% exact-span absence and Hit@1 0.123. When coverage finds *split* spans, adjust chunking or overlap. When it finds *absent* spans, inspect the parser output and switch parsers if required content is genuinely missing.

The end-to-end check supports the top decision but not the full ranking. Prod leads answer accuracy among three parsers, while the near-tied lower pair reverses. The training evidence is also limited: neither RADP method meets the pilot target, the retrieval-reward variants do not separate from the matched fidelity control, and SimPO has negative point estimates. Lower TextNED at the DPO checkpoints is only a post hoc association. Incomplete boundary evidence prevents a causal explanation (Appendix G). In practice, use retrieval to select the pipeline, coverage to locate loss, and training only after both steps.

Limitations

Data scope and frames. The reported denominators are not interchangeable. KoGovDoc-RAG selection indexes 294 pages for 663 Q–A, whereas training indexes only the 242 evidence-bearing pages. The pilot uses 73 pages and 202 Q–A. The complete parser comparison contains five 294-page outputs. Its BC correlation uses the four outputs with defined BC ($r = -0.74$). The $r = -0.83$ estimate additionally includes Marker's partial 38-page output. OHR-Bench uses 1,043 Law–Manual Q–A for perturbations and 2,036 Q–A for training compatibility. The Law–Manual frame contains dependent perturbations and does not establish domain generality. We exclude the mixed-version seven-domain artifacts (Appendix A).

MinerU audit. MinerU-off disables table recognition and remains only as a submitted-output diagnostic. In the later MinerU-on run, table-evidence absence falls from 87.9% to 41.7%. Prod's rate is 13.9%. MinerU-on still reaches only 0.137 RCPS and 0.123 Hit@1, compared with Prod's 0.583 and 0.549. The two MinerU runs have similar BC (0.713 and 0.716), but they also differ in software and retrieval environments. Their differences therefore do not isolate the effect of table recognition. Domain results also vary: MinerU-on RCPS is 0.046 on government pages and 0.486 on arXiv pages.

Reference, Q–A, and label quality. KoGovDoc-RAG uses manually de-noised Qwen3-VL-30B pseudo-references and LLM-generated Q–A. Neither complete set was human-verified. Two 100-case checks address narrower questions. An LLM check accepted 94/100 sampled Q–A. A blind two-author study evaluates coverage labels. The fixed probe controls query variation but not possible Qwen-lineage bias. OHR-Bench supplies external evidence, although the six-domain result is a compatibility analysis rather than a full v2 rerun. RCPS evaluates verbatim spans, not implicit or paraphrased answers. Coverage also depends on its matcher: GPT-5.4 reclassifies 56% of Prod's exact-match-absent cases as surface artifacts. Finally, the same GPT-5.4 checkpoint generates and judges end-to-end answers, so that experiment supports only the top parser choice.

Training. The KoGovDoc-RAG R2 result is exploratory. Its +1.96-point Hit@5 estimate (95% CI $[-1.06, +5.03]$) uses 242 pages and a configuration selected after several settings were inspected. On the separate 2,036-Q–A OHR compatibility subset, R2, R3, and RADP-Distill exceed Prod by 0.95, 1.15, and 1.36 Hit@5 points. Distill-minus-R2 is +0.41 points (95% CI $[-0.43, +1.26]$), and Distill-minus-R3 is +0.21 points (95% CI $[-0.61, +1.05]$). This subset was defined after a version audit and is not a full v2 rerun. We do not adjust for multiple checkpoints and secondary metrics, and the bootstrap resamples Q–A rather than shared evidence pages. These are post-audit estimates, not the original confirmatory analysis.

Scope. RCPS targets answer-span retrieval rather than generated answers. The end-to-end experiment checks only three parsers. The preference candidates come from Prod, so another parser pool may behave differently. We do not test token-level reinforcement learning, curricula, or chunk-level oracle distillation. Broader parser pools and training of chunkers, embedders, and rerankers remain future work.

References

Shuai Bai, Yuxuan Cai, Ruizhe Chen, Keqin Chen, Xionghui Chen, Zesen Cheng, Lianghao Deng, Wei Ding, Chang Gao, Chunjiang Ge, Wenbin Ge, Zhifang Guo, Qidong Huang, Jie Huang, Fei Huang, Binyuan Hui, Shutong Jiang, Zhaohai Li, Mingsheng Li, and 45 others. 2025. Qwen3-VL technical report. Preprint, arXiv:2511.21631.

Jianlv Chen, Shitao Xiao, Peitian Zhang, Kun Luo, Defu Lian, and Zheng Liu. 2024. M3-Embedding: Multi-linguality, multi-functionality, multi-granularity text embeddings through self-knowledge distillation. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 2318–2335, Bangkok, Thailand. Association for Computational Linguistics.

Yew Ken Chia, Liying Cheng, Hou Pong Chan, Maojia Song, Chaoqun Liu, Mahani Aljunied, Soujanya Poria, and Lidong Bing. 2025. M-LongDoc: A benchmark for multimodal super-long document understanding and a retrieval-aware tuning framework. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 9233–9250, Suzhou, China. Association for Computational Linguistics.

Max Conti, Manuel Faysse, Gautier Viaud, Antoine Bosselut, Céline Hudelot, and Pierre Colombo. 2025. Context is gold to find the gold passage: Evaluating and training contextual document embeddings. In *Proceedings of the*

2025 Conference on Empirical Methods in Natural Language Processing, pages 22594–22608, Suzhou, China. Association for Computational Linguistics.

André V. Duarte, João DS Marques, Miguel Graça, Miguel Freire, Lei Li, and Arlindo L. Oliveira. 2024. LumberChunker: Long-form narrative document segmentation. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 6473–6486, Miami, Florida, USA. Association for Computational Linguistics.

Manuel Faysse, Hugues Sibille, Tony Wu, Bilel Omrani, Gautier Viaud, Céline Hudelot, and Pierre Colombo. 2025. ColPali: Efficient document retrieval with vision language models. In *International Conference on Learning Representations*.

Michael Günther, Isabelle Mohr, Daniel James Williams, Bo Wang, and Han Xiao. 2024. Late chunking: Contextual chunk embeddings using long-context embedding models. Preprint, arXiv:2409.04701.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*.

Yu Meng, Mengzhou Xia, and Danqi Chen. 2024. SimPO: Simple preference optimization with a reference-free reward. In *Advances in Neural Information Processing Systems*, volume 37.

Thang Nguyen, Peter Chin, and Yu-Wing Tai. 2024. Reward-RAG: Enhancing RAG with reward driven supervision. Preprint, arXiv:2410.03780.

Junbo Niu, Zheng Liu, Zhuangcheng Gu, Bin Wang, Linke Ouyang, Zhiyuan Zhao, Tao Chu, Tianyao He, Fan Wu, Qintong Zhang, Zhenjiang Jin, Guang Liang, Rui Zhang, Wenzheng Zhang, Yuan Qu, Zhifei Ren, Yuefeng Sun, Zirui Tang, Boyu Niu, and 40 others. 2026. MinerU2.5: A decoupled vision-language model for efficient high-resolution document parsing. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 6: Industry Track)*, pages 13–42, San Diego, California, USA. Association for Computational Linguistics.

Linke Ouyang, Yuan Qu, Hongbin Zhou, Jiawei Zhu, Rui Zhang, Qunshu Lin, Bin Wang, Zhiyuan Zhao, Man Jiang, Xiaomeng Zhao, Jin Shi, Fan Wu, Pei Chu, Minghao Liu, Zhenxiang Li, Chao Xu, Bo Zhang, Botian Shi, Zhongying Tu, and Conghui He. 2025. OmniDocBench: Benchmarking diverse PDF document parsing with comprehensive annotations. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 24838–24848.

Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn. 2023. Direct preference optimization: Your language model is secretly a reward model. In *Advances in Neural Information Processing Systems*, volume 36.

Ishneet Sukhvinder Singh, Ritvik Aggarwal, Ibrahim Allahverdiyev, Muhammad Taha, Aslihan Akalin, Kevin Zhu, and Sean O'Brien. 2024. ChunkRAG: Novel LLM-chunk filtering method for RAG systems. Preprint, arXiv:2410.19572.

Saurabh K. Singh and Sachin Raj. 2026. Benchmarking complex multimodal document processing pipelines: A unified evaluation framework for enterprise AI. Preprint, arXiv:2604.26382.

Minchae Song. 2026. Defining the problem: The impact of OCR quality on retrieval-augmented generation performance and strategies for improvement. *Information Processing & Management*, 63(1):104368.

Lin Sun, Wangdexian, Jingang Huang, Linglin Zhang, Change Jia, Zhengwei Cheng, and Xiangzheng Zhang. 2026. When good OCR is not enough: Benchmarking OCR robustness for retrieval-augmented generation. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 6: Industry Track)*, pages 884–894, San Diego, California, USA. Association for Computational Linguistics.

Baode Wang, Biao Wu, Weizhen Li, Meng Fang, Zuming Huang, Jun Huang, Yanjie Liang, Haozhe Wang, Ling Chen, Wei Chu, and Yuan Qi. 2026a. Infinity-Parser: Layout-aware reinforcement learning with high-quality document parsing dataset. In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 1647–1667, San Diego, California, United States. Association for Computational Linguistics.

Liang Wang, Nan Yang, Xiaolong Huang, Linjun Yang, Rangan Majumder, and Furu Wei. 2024. Multilingual E5 text embeddings: A technical report. Preprint, arXiv:2402.05672.

Zhengren Wang, Dongsheng Ma, Huaping Zhong, Jiayu Li, Wentao Zhang, Bin Wang, and Conghui He. 2026b. AgenticOCR: Parsing only what you need for efficient retrieval-augmented generation. Preprint, arXiv:2602.24134.

Shi-Qi Yan, Quan Liu, and Zhen-Hua Ling. 2025. RPO: Retrieval preference optimization for robust retrieval-augmented generation. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume*

1: Long Papers), pages 5228–5240, Vienna, Austria. Association for Computational Linguistics.

Junyuan Zhang, Qintong Zhang, Bin Wang, Linke Ouyang, Zichen Wen, Ying Li, Ka-Ho Chow, Conghui He, and Wentao Zhang. 2025a. OCR hinders RAG: Evaluating the cascading impact of OCR on retrieval-augmented generation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 17443–17453.

Yanzhao Zhang, Mingxin Li, Dingkun Long, Xin Zhang, Huan Lin, Baosong Yang, Pengjun Xie, An Yang, Dayiheng Liu, Junyang Lin, Fei Huang, and Jingren Zhou. 2025b. Qwen3 Embedding: Advancing text embedding and reranking through foundation models. Preprint, arXiv:2506.05176.

Jihao Zhao, Zhiyuan Ji, Zhaoxin Fan, Hanyu Wang, Simin Niu, Bo Tang, Feiyu Xiong, and Zhiyu Li. 2025a. MoC: Mixtures of text chunking learners for retrieval-augmented generation system. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 5172–5189, Vienna, Austria. Association for Computational Linguistics.

Jihao Zhao, Zhiyuan Ji, Yuchen Feng, Pengnian Qi, Simin Niu, Bo Tang, Feiyu Xiong, and Zhiyu Li. 2024. Meta-chunking: Learning text segmentation and semantic completion via logical perception. Preprint, arXiv:2410.12788.

Yao Zhao, Yantian Ding, Zhiyue Zhang, Dapeng Yao, and Yanxun Xu. 2025b. LMAR: Language model augmented retriever for domain-specific knowledge indexing. Preprint, arXiv:2508.05672.

Appendix roadmap. Appendices A–D support the intrinsic-metric, RCPS-selection, and coverage analyses (C1–C3). Appendices E–G document the secondary parser-training study (C4), and Appendix H records dataset and release scope.

A OHR-Bench Alignment and Noise Sensitivity

We use two OHR-Bench evaluation frames because the initial seven-domain artifacts mixed benchmark versions. Those artifacts paired Q–A from the legacy 4,330-page release with parser outputs from the current 8,561-page bundle. For 223 legacy Q–A assigned to the `notes` domain, the old evaluation mapped them to v2 `administration`. The corresponding evidence is instead under v2 `textbook`. As a result, the lookup selected unrelated documents and stored zeros. One further legacy textbook page, referenced by five Q–A, is absent from v2. After excluding these 228 affected cases, we retain a source-aligned compatibility subset of 2,036 Q–A across six domains. This subset supports compatibility checks, not a full v2 rerun. We therefore discard the correlation computed from the mixed-version frame.

The Law–Manual files retain matching document and page identifiers across both releases. We use this 1,043-Q–A frame for the C1 perturbation study. Its 15 variants comprise three benchmark outputs (reference, MinerU, and Qwen2.5-VL), three formatting perturbations, and nine semantic perturbations. The semantic variants apply three noise severities to MinerU, Qwen2.5-VL, and GOT; no unperturbed GOT output is available. Across the 15 variants, the descriptive BC–RCPS correlation is $r = -0.35$. This value should not be interpreted as a correlation across 15 independent parsers because variants within a family share the same source output. Table 2 reports the semantic families on the same Law–Manual pages with the same three-retriever average.

Family	Metric	Clean	Mild	Mod.	Severe
MinerU	BC	0.657	0.628	0.651	0.631
	RCPS	0.595	0.476	0.384	0.265
Qwen2.5-VL	BC	0.563	0.563	0.563	0.564
	RCPS	0.545	0.534	0.523	0.497
GOT	BC	—	0.586	0.620	0.624
	RCPS	—	0.461	0.385	0.298

Table 2: Semantic noise lowers RCPS within every family, while BC is stable, non-monotonic, or increasing. Results use 1,043 source-aligned Law–Manual Q–A. Clean denotes unperturbed output. Mild, Mod., and Severe denote increasing semantic noise. GOT has no clean output.

B Does the RCPS Top Choice Also Lead in Answer Generation?

We ask whether the RCPS top choice among Prod, MinerU-on, and PaddleOCR also leads in answer generation. The evaluation covers all 663 KoGovDoc-RAG Q–A. BGE-M3 first retrieves the top five chunks from each parser corpus. `gpt-5.4-2026-03-05` then generates an answer from those chunks and judges it against the reference answer.

This generation pipeline is not identical to the main RCPS protocol. The RCPS values in Table 3 average three retrievers at three retrieval depths, whereas answer generation uses only BGE-M3 top-five retrieval. Each row nevertheless uses the same parser output for its RCPS and generation results. In particular, the MinerU row uses MinerU-on. MinerU-off scores 0.212 RCPS, but it has no corresponding answer-generation run and is therefore excluded from this comparison.

Prod achieves 72.5% answer accuracy, compared with 23.8% for MinerU-on and 20.5% for PaddleOCR. Thus, the RCPS top choice also leads in judged answer accuracy. The nearly tied lower pair reverses, so this experiment does not validate their relative order. Moreover, the same model checkpoint generates and judges the answers. We therefore treat the experiment only as an end-to-end check of the top choice, not as validation of the full parser ranking.

C Is the Absent Label Robust?

This appendix assesses the exact-span absent label with alternative matchers, a cross-family judge, and a parser-masked human check.

Parser	Ans. acc.	EM	Answered	RCPS
Prod	**72.5**	**49.8**	87.9	0.583
MinerU (on)	23.8	14.5	39.1	0.137
PaddleOCR	20.5	11.5	45.9	0.140

Table 3: The RCPS top choice, Prod, also has the highest answer accuracy in this three-parser check. The near-tied lower pair reverses. Generation uses BGE-M3 top-five retrieval over 663 Q–A, with MinerU-on used for both RCPS and generation. RCPS averages the main three-retriever grid at three retrieval depths. EM is normalised-span exact match. Answered excludes `NO ANSWER`. Ans. acc., EM, and Answered are percentages. RCPS uses a 0–1 scale. The same GPT-5.4 checkpoint generates and judges answers, so this checks only the top choice.

Parser I/O and the "absent" label. Each PDF page is first rendered as an image. A document parser maps that image to a linear Markdown transcription. The intended output preserves body text in reading order, ATX (#-style) headings, and GitHub-flavoured Markdown tables. It discards layout coordinates, fonts, figure imagery, and page furniture unless they are transcribed as text. The input collection includes born-digital and scanned PDFs, multi-column layouts, merged table cells, stamps, seals, and text embedded in figures.

Only the resulting Markdown is chunked, embedded, and retrieved. Under the operational definition in §3.2, a reference answer is *absent* when its normalised exact span has no match in the source-page transcription. The reference Markdown used for TextNED and the reference answer spans come from manually de-noised Qwen3-VL-30B output. We therefore treat both as pseudo-ground truth.

Worked example: a corrupted table unit. On KoGov page `val_0155`, the pseudo-reference and parser output differ as follows:

Pseudo-reference answer	$A = 180\,\mathrm{m}^2$
MinerU-on output	`A = 180m`

MinerU-on omits the exponent. The shared normalisation removes formatting, but it does not treat m^2 and m as equivalent. The complete answer is therefore absent under the exact-span relevance rule. Rechunking cannot reconstruct the missing exponent.

Alternative matchers. The reference Markdown, reference answers, and Prod share Qwen3 lineage. To test whether this shared lineage explains the absent-rate gap, we apply five model-free presence matchers to all 663 Q–A. L0 uses raw substring matching. L1 applies Unicode Normalization Form KC (NFKC), case folding, whitespace normalisation, and Markdown normalisation. L2 additionally removes digit separators and in-content punctuation. L3 requires at least 90% order-free token recall. L4 requires the longest common substring to cover at least 80% of the answer. L3 and L4 are alternative tolerant criteria, not successive steps in a nested ladder. If surface form explains most of the gap, at least one tolerant matcher should narrow it.

Absent %	L0	L1	L2	L3	L4
Prod (2B)	24.1	20.2	19.6	24.1	16.9
30B teacher	22.8	18.9	16.6	19.2	16.3
2B base	30.0	25.2	24.7	27.0	21.1
MinerU-off	74.8	70.4	68.0	73.0	68.6
PaddleOCR	67.0	62.7	62.0	59.9	60.5
MinerU−Prod	+50.7	+50.2	+48.4	+48.9	+51.7
PaddleOCR−Prod	+42.8	+42.5	+42.4	+35.7	+43.6

Table 4: The OCR-parser absence gaps over Prod persist across five presence matchers (663 Q–A, no retriever; MinerU-off). L0 is raw matching; L1 is normalised exact matching; L2 removes additional punctuation; L3 uses token recall; and L4 uses longest-common-substring coverage. The two other Qwen3-derived parsers remain within 5.9 percentage points of Prod.

Cross-family model judge. We next ask GPT-5.4, a different model family from the evaluated parsers, whether evidence remains recoverable in each L1-absent parser–answer pair (Table 5). *Present* denotes a recoverable surface mismatch. *Degraded* means that evidence remains but is too corrupted for reliable retrieval, and *absent* denotes missing content. We combine the latter two labels as *retrieval-unusable*. We apply the same procedure independently to every parser. The MinerU analysis here uses MinerU-off.

The Q–A pairs were also generated by a GPT-family model, so this judge is not independent of Q–A generation. The model-free matchers above avoid a GPT-family matcher, but they still depend on the generated Q–

A. The judge marks 56% of Prod's L1-absent cases as surface artifacts, compared with about 16% for both MinerU and PaddleOCR. After those artifacts are removed, the MinerU–Prod retrieval-unusable gap remains 50.4 percentage points. The corresponding gaps are 50.2 points under L1 exact matching and 51.7 points under the L4 character-tolerant criterion.

	present (artifact)	degraded	absent	retrieval-unusable
Prod	56%	2%	42%	**8.9%**
MinerU-off	16%	9%	75%	**59.3%**
PaddleOCR	16%	22%	63%	**52.8%**

Table 5: A cross-family judge preserves the OCR-parser gap after recoverable surface mismatches are separated. GPT-5.4 labels 1,017 parser–answer cases marked absent by L1 (134 Prod, 467 MinerU-off, and 416 PaddleOCR). Degraded and absent are retrieval-unusable. The first three columns are percentages within each parser's L1-absent set. The final column is a percentage of all 663 Q–A.

Human verification of absent labels. Two authors independently labelled 100 parser-masked L1-absent cases, stratified by question type. The sample contains 50 MinerU-on, 30 Prod, and 20 PaddleOCR cases. This study evaluates coverage labels. It is separate from the 100-pair Q–A quality check. Before adjudication, the authors agreed on 81/100 cases, with $\kappa = 0.615$. They then jointly resolved the remaining 19 cases.

After adjudication, 42/50 MinerU-on cases were retrieval-unusable (84.0%, Wilson 95% CI $[71.5, 91.7]$). For Prod, 12/30 cases were retrieval-unusable (40.0%, $[24.6, 57.7]$). For PaddleOCR, the rate was 19/20 (95.0%, $[76.4, 99.1]$).

The automated-judge set contains counterparts for 93 human-labelled cases: 43 MinerU, 30 Prod, and 20 PaddleOCR. The seven excluded cases are MinerU-on examples with no MinerU-off counterpart. Across the full automated-judge sets, retrieval-unusable rates are 84.2% for MinerU-off, 84.1% for PaddleOCR, and 44.0% for Prod. Each falls within the corresponding human-sample Wilson interval. Among the 93 cases shared by the human and automated evaluations, pooled binary agreement is 84/93 (90.3%). The human labels thus confirm the direction of the OCR-parser gap in this stratified sample. Different sampling fractions and MinerU configurations preclude a population-level replication claim. The sampling manifest, both rating files, and the 19-case adjudication record were verified with the scoring script and are retained in an author-only audit package.

Causes of absence. For Prod, 23/165 table-evidence answers are exact-match absent (13.9%). The count is 43/201 (21.4%) for factoid answers and 63/290 (21.7%) for procedural answers. For figure-evidence answers, 5/7 are absent (71.4%), but this seven-case estimate should not be generalised. This high point estimate is consistent with figure imagery not being transcribed into retrievable text.

We observe three recurring causes. First, parsers drop table cells before the content reaches Markdown. For table-evidence answers, absent rates are 87.9% for MinerU-off, 41.7% for MinerU-on, and 13.9% for Prod. Second, parsers skip text embedded in captions, stamps, seals, or figure labels. Third, they misrecognise numerals or units beyond the L2/L4 tolerance.

D Chunkers Rarely Split Exact Spans

We apply the retriever-free coverage diagnostic to Prod output for all 663 Q–A over 294 pages. As defined in §3.2, text matching labels each reference span as covered, split, or absent. Because the parser output is fixed, the pre-chunking absent rate remains 20.2% across all eight chunkers. No tested chunker splits more than 2.3% of exact answer spans.

Chunker	covered	split	absent
md_h3	79.8%	0.0%	20.2%
md_h2	79.8%	0.0%	20.2%
md_h1	79.8%	0.0%	20.2%
parser_native	78.1%	1.7%	20.2%
fixed500	77.5%	2.3%	20.2%
fixed500_ov200	79.8%	0.0%	20.2%
fixed1000	79.3%	0.5%	20.2%
fixed1000_ov200	79.8%	0.0%	20.2%

Table 6: No tested chunker splits more than 2.3% of reference spans. The pre-chunking absent rate stays 20.2% (Prod, 294 pages, 663 Q–A). `parser_native` uses blank-line paragraph boundaries. md_hk uses Markdown heading depth k. fixed N[_ov$M$] uses N-character windows with optional M-character overlap.

E How RADP-DPO Builds Preferences (C4)

Common training setup. All preference-based variants are Prod LoRA adapters ($r = 8$, $\alpha = 32$) (Hu et al., 2022). The RADP-aux sweep instead trains LoRA adapters from Qwen3-VL-2B-Instruct and applies them to Prod for evaluation, matching the executed pipeline. Evaluation uses deterministic decoding with temperature 0 and a 1,536-token limit. Candidate generation for preference construction is stochastic. The 2,667-page training corpus, which supplies 6,164 generated Q–A, is page-disjoint from every evaluation frame. RADP-aux sweeps $\lambda \in \{0, 0.1, 0.3, 0.5\}$ for its contrastive loss. Unless stated otherwise, confidence intervals use 1,000 paired Q–A-level percentile-bootstrap resamples.

Preference construction. RADP-DPO samples alternative Prod outputs for each page in the separate training corpus. It chunks every candidate parse and scores those chunks against questions linked to the page, with other-page chunks as distractors. The page-local reward is BGE-M3 MRR averaged over $k \in \{1, 5, 10\}$. This training reward differs from the evaluation RCPS, which averages three retrievers. Multilingual-e5-large and Qwen3-Embedding-8B are therefore unseen by the reward.

Candidate parses that differ by at least 5 percentage points in page-local reward form a DPO pair. The higher-scoring parse is the chosen example. R1 samples $K = 2$ parses with uniform distractors. R2 starts from the R1 checkpoint and trains on a second preference round with $\beta = 0.1$. R3 expands the candidate pool to $K = 16$ by adding 14 parses to the two R1 candidates. It also adds full-corpus hard negatives: chunks from other pages that are closest to the current queries.

Training objective. Let x be the input page, and let y^+ and y^- be its chosen and rejected parses. Define $\Delta_\theta = \log \pi_\theta(y^+ \mid x) - \log \pi_\theta(y^- \mid x)$. Define $\Delta_{\text{ref}} = \log \pi_{\text{ref}}(y^+ \mid x) - \log \pi_{\text{ref}}(y^- \mid x)$. During training, π_θ enables the trainable LoRA adapters, whereas π_{ref} disables them:

$$\mathcal{L}_{\text{DPO}} = -\log \sigma(\beta[\Delta_\theta - \Delta_{\text{ref}}]). \tag{2}$$

The SimPO control uses a length-normalised chosen–rejected log-probability margin ($\beta = 2.0$, target margin $\gamma = 1.0$) without a reference policy.

F Parser-Training Results Across Frames (C4)

Across evaluation frames, the observed training differences are small or uncertain. The 73-page pilot misses its target. Every DPO and SimPO interval in the 242-page KoGovDoc-RAG analysis crosses zero. On the OHR-Bench compatibility subset, the R2, R3, and RADP-Distill Hit@5 estimates are about one percentage point above Prod.

Evaluation frames. Before the pilot, success required at least a 5-percentage-point RCPS gain and a 95% CI lower bound above zero. The pilot held out 73 pages and 202 Q–A from 242 evidence-bearing pages. Its best RADP-aux and RADP-DPO estimates missed that target, and their intervals crossed zero.

The later KoGovDoc-RAG analysis pools the 169 development pages with the 73 held-out pages. It is therefore not an independent holdout. All 663 Q–A have evidence on these 242 indexed pages. The 294-page selection index additionally contains 52 Q–A-free distractor pages. The cross-domain analysis instead uses the 2,036-Q–A, six-domain OHR-Bench compatibility subset in Table 7. All training examples come from the separate 2,667-page corpus. The preference-based systems are Prod LoRA adapters; RADP-aux uses the cross-base training and evaluation setup described above.

KoGovDoc-RAG results. The three RADP-DPO milestones have parser-native Hit@5 point estimates 1.96–2.11 percentage points above Prod. For each checkpoint, the paired-bootstrap proportion $P_{\text{boot}}[\Delta\text{Hit@5} > 0]$ is about 0.90 (Table 8). This descriptive bootstrap proportion is not a p-value. Every two-sided interval crosses zero at $n = 663$.

These estimates are exploratory. The reporting configuration was selected after examining multiple chunker, retriever, and retrieval-depth settings, without a family-wise multiplicity correction. The bootstrap also re-

Δ vs Prod (pp)	Hit@1	Hit@5	Hit@10	MRR@10	nDCG@5
RADP-DPO-R2	+0.59	+0.95	+0.90	+0.78	+0.82
RADP-DPO-R3	+1.46	+1.15	+0.90	+1.30	+1.28
RADP-Distill	+0.98	+1.36	+1.47	+1.12	+1.16

Table 7: R2, R3, and RADP-Distill have small positive deltas on the audited OHR-Bench compatibility subset, but this is not a full v2 rerun. The subset contains 2,036 source-aligned Q–A across six domains. It uses a three-retriever average with 1,000 Q–A-level paired-bootstrap resamples. Deltas are percentage points. Hit@5 intervals are R2 $[+0.33, +1.54]$, R3 $[+0.31, +2.05]$, and Distill $[+0.43, +2.29]$.

Variant (vs Prod = 0.6757)	Hit@5	ΔHit@5 [95% CI]	$P_{\text{boot}}[\Delta\text{Hit@5}>0]$	ΔRCPS (pp)
RADP-DPO-R3	0.6968	+2.11 $[-0.90, +5.13]$	0.91	+1.72
RADP-DPO-R1	0.6963	+2.06 $[-0.96, +5.13]$	0.91	+0.57
RADP-DPO-R2	0.6953	+1.96 $[-1.06, +5.03]$	0.90	+0.47
RADP-SimPO (ctrl)	0.6687	−0.70 $[-3.77, +2.31]$	0.32	−1.56

Table 8: All KoGovDoc-RAG DPO point estimates are positive, but every two-sided interval crosses zero. The SimPO point estimate is negative. The analysis indexes 242 pages for 663 Q–A and uses parser-native chunking with 10,000 Q–A-level paired-bootstrap resamples. Hit@5 averages three retrievers. Prod is 0.6757. $P_{\text{boot}}[\Delta\text{Hit@5} > 0]$ is a bootstrap proportion, not a p-value. Across three independent R1-recipe seeds, the standard deviation of mean ΔHit@5 is 0.90 percentage points.

samples Q–A pairs rather than evidence pages. Table 7 omits R1 because no source-aligned cross-domain array is available for that checkpoint.

RADP-aux does not meet the pre-specified criterion. On the held-out 73-page, 202-Q–A fold, $\lambda = 0.1$ gives the largest estimate relative to the $\lambda = 0$ fine-tuning control. The gain is +1.13 RCPS percentage points with md-h3 and +2.31 with parser-native chunking. Estimates decline at larger values of λ, and every paired confidence interval relative to $\lambda = 0$ includes zero.

SimPO has negative point estimates. Relative to Prod on the 242-page fold, its three-retriever mean Hit@5 difference is −0.85 percentage points with md-h3 and −0.70 with parser-native chunking. Both confidence intervals cross zero. These results do not isolate whether reference removal, length normalisation, or another optimisation difference causes the change.

Matched fidelity control. RADP-Distill changes only the pair-selection signal: it ranks candidates by edit distance rather than retrieval. On the same 2,036 observations, its Hit@5 estimate is 1.36 points above Prod. Its direct differences from R2 and R3 are +0.41 and +0.21 points, with both paired intervals crossing zero. The experiment therefore does not show that retrieval-reward pair selection outperforms the fidelity control.

G Do Training Point Estimates Track Fidelity or Boundaries? (C4)

The available measurements do not identify why the training point estimates change. DPO checkpoints have lower TextNED than Prod. However, TextNED does not reproduce their retrieval order, and the boundary evidence is incomplete. Table 9 reports Boundary Clarity (BC), normalised edit distance to reference Markdown (TextNED), and chunking shape on the 242-page corpus. TextNED is the character-level Levenshtein distance divided by the longer string length.

Variant	len	c/pg	clen	BC↑	TextNED↓
Prod (baseline)	1385	4.82	274	0.630	0.240
aux λ=0.0	799	2.96	245	0.553	0.626
aux λ=0.1	1380	4.05	321	0.652	0.423
aux λ=0.3	1930	6.73	272	0.470	0.332
aux λ=0.5	2035	5.82	327	0.556	0.318
DPO-R1	1606	4.97	311	0.646	0.167
DPO-R2	1610	4.83	321	0.647	**0.163**
DPO-R3	1514	4.88	300	—	0.185
SimPO	1703	5.31	305	0.601	0.186

Table 9: DPO checkpoints have lower TextNED than Prod on the 242-page corpus, but the metric does not reproduce their retrieval order. Len is mean parsed characters per page. c/pg is mean chunks per page, and clen is mean characters per chunk. BC was not computed for R3.

DPO checkpoints have lower TextNED. The three RADP-DPO checkpoints have TextNED values of 0.163–0.185, compared with Prod's 0.240. Relative to the pseudo-reference, these distances are 23–32% lower. RADP-aux instead has TextNED values of 0.318–0.626.

All three DPO checkpoints also have positive Hit@5 point estimates in Table 8. However, TextNED does not order them. R2 has the lowest TextNED and the smallest DPO Hit@5 difference. R3 has the highest TextNED among the DPO checkpoints and the largest Hit@5 difference. The co-occurrence is consistent with fidelity as one possible explanation, but these observational results do not identify a causal mechanism.

Available structure evidence is incomplete. R1 and R2 have 4.97 and 4.83 chunks per page, close to Prod's 4.82. Their mean chunk lengths, however, increase from 274 characters for Prod to 311 and 321. Their BC values are 0.646 and 0.647, compared with Prod's 0.630. R3 has comparable chunking shape, but no reproduced BC value. The missing R3 value and the lack of uncertainty estimates prevent a systematic conclusion about boundary change.

Interpretation. One possible explanation for RADP-aux's smaller point estimates is that its hidden-state objective can affect retrieval only indirectly through generated Markdown. The aligned RADP-Distill retrieval comparison does not separate the two pair-selection objectives. Previously computed OHR TextNED breakdowns use artifacts that are not aligned with the audited subset, so we exclude them from the mechanism analysis.

H Dataset Composition and Release Status

Dataset composition. KoGovDoc-RAG contains 229 Korean government pages and 65 arXiv pages. The two sources contribute 527 and 136 verbatim-answerable Q–A, respectively. The separate 2,667-page training set is page-disjoint from all 294 evaluation pages. Prod exceeds its 2B base by 4.9 Hit@1 points, whereas the audited Prod–MinerU-on difference is 42.6 points. This comparison provides scale, not a formal bound on fine-tuning effects.

Stored retriever ablation. This ablation predates the MinerU-on audit and uses the submitted MinerU-off grid. Relative to full RCPS, BGE-M3 alone changes only the near-tied 30B–Prod order ($\tau = 0.80$). Lower parser ranks and the chunker order remain unchanged. It provides no evidence about the MinerU-on–PaddleOCR order.

Artifact status. The repository contains the frozen probe and page map, both MinerU configurations' 294-page outputs, retrieval and diagnostic code, aligned per-Q–A arrays and stability results, and deterministic OHR-compatibility and MinerU-output audits. A separate public release provides all nine evaluated LoRA adapters, with portable configurations, hashes, and available trainer states. The adjudicated human-study inputs remain in an author-only audit package; the public repository reports their aggregate results. We verified the public checkpoint download and CPU-only artifact gates from a clean checkout. A full end-to-end rerun still requires external source documents, parser outputs, and embedding caches, and we do not claim a full OHR-Bench v2 rerun.